AI FAQ ASSISTANT - PROJECT REPORT

Phase 3: Project Design Phase

System Architecture

The API follows a layered architecture:
• Clients: Web app, mobile app, Postman
• Gateway: Express app, CORS, JSON parser
• Middleware: Authentication, validation, error handling
• Controllers: Coordinate requests and shape responses
• Services: Gemini AI service
• Data: MongoDB with Mongoose schemas

MVC Pattern Implementation

• Model: Mongoose schemas with validation rules
• View: REST routes mapping to controllers
• Controller: Request processing and JSON responses

Data Model

Two main collections:
1. User: _id, name, email, password (hashed), role, timestamps
2. FAQ: _id, question, answer, category, createdBy (reference), timestamps
Logical entities: AI Generation, Category

Relationships

• User-FAQ: One-to-many
• Category-FAQ: One-to-many
• User-AI Generated FAQ: One-to-many
• FAQ-Search Results: Logical relationship

Security Design

• Password hashing with bcrypt (10 rounds)
• JWT token-based authentication
• Author-based authorization
• Environment variables for secrets
• AI endpoints protected routes

- Centralized error handling